

Jiawen Kang

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ERB 503
The Chinese University of Hong Kong
Shatin, N.T.
Hong Kong SAR

PARTICULARS

EDUCATION

The Chinese University of Hong Kong (CUHK)
Ph. D. in Information System
Thesis Advisor: [Prof. Helen Meng](#)

Hong Kong SAR, China
Sep. 2021 - Feb. 2027 (Exp.)

Jilin University
B. Eng. in Electrical Engineering

Changchun, China
Sep. 2015 - Jul. 2019

RESEARCH INTERESTS

I build speech and language models for human conversation, with a current focus on agentic systems for clinical conversational agents. Previously, my work centered on general speech understanding, including speech and speaker recognition.

RESEARCH HIGHLIGHTS

- Recipient of the **2025 IEEE Ganesh N. Ramaswamy Memorial Student Grant**, an annual honor recognizing one outstanding student paper globally in speech and language processing
- Awarded the **ISCA Award for the Best Paper Published in Speech Communication** (2019–2023); selected as the sole best paper among all journal publications over a 5-year span (co-first author)
- Received the **CUHK Outstanding Student Award** (1 of 26 postgraduates university-wide) for pioneering research on conversational spoken language processing
- Authored 11 papers as first or co-first author in top-tier spoken language processing venues, including 2 currently under review

RESEARCH EXPERIENCES

- **Tencent AI Lab & Tencent Hunyuan**
Full-time Research Intern *Aug. 2025 - Feb. 2026*
Post-training for large audio language models
- **The Chinese University of Hong Kong**
Full-time Research Assistant, Advisor: [Prof. Helen Meng](#) *Nov. 2020 - Jul. 2021*
Spoken dialogue agents with backchannel generation

- **Tsinghua University**
Full-time Research Assistant, CSLT, Advisor: [Prof. Dong Wang](#) Sep. 2019 - Oct. 2020
Robust speaker recognition
- **Jilin University**
Research Intern, College of Instrumentation and Electrical Engineering Nov. 2017 - Nov. 2018
Denoising algorithms for fiber-optical sensing

RESEARCH TOPICS

Conversational Diagnostic AI

Ph.D. Thesis Research, funded by RGC Theme-Based Research (T45-407/19N) 2022 - Now

- **Agentic RL** - RLVR post-training agent for multi-turn multi-task clinical dialogue [2,4]
- **Heat Diffusion Graph** - Heat-diffusion framework for memory retrieval sequential trajectory [3]
- **Statistical Robustness** - Look into within-class variation issue in LM representations [5]
- **Narrative Analysis** - Cross-modality alignment method for narrative analysis [11]
- **Clinical Coordination** - Managed a multi-year clinical conversation study across data collection and medical-team collaboration, establishing a 2000+ hour internal corpus from 1000+ participants
- Resulted in the CUHK Outstanding Student Award

Multi-talker Speech Recognition for Clinical Conversations

Ph.D. Thesis Research 2023 - 2024

- **Audio Disentanglement** - Bayes-risk regularized speaker-aware CTC to induce speaker-content disentanglement [6]
- **Network Architecture** - Task-prior-guided network architecture for multi-talker recognition [8]
- Recognized by the 2025 IEEE Ganesh N. Ramaswamy Memorial Student Grant

Large Audio Language Models

Research Intern, Tencent AI Lab & Hunyuan Aug. 2025 - Feb. 2026

- **Post-Training** - SFT post-training for ASR and paralinguistic understanding in Covo-Audio, a 7B open-sourced audio LLM; achieving SOTA performance
- **Evaluation Infrastructure** - Developed an evaluation platform for standardized model benchmarking and performance tracking across model variants
- Designated as a Core Contributor of Covo-Audio [Technical Report]

Robust Speaker Recognition

Tsinghua University 2019 - 2020

- **Benchmark Establishment** - Foundation of CN-Celeb, a premier and widely adopted in-the-wild speaker recognition benchmark [10, 21]
- **Meta Learning** - First method for domain-invariant speaker embeddings, inspired by model-agnostic meta-learning [9]
- Resulted in the Best Paper Published in Speech Communication (2019–2023)

OTHER PROJECTS

- **Detection of Neurocognitive Disorder with Structural Magnetic Resonance Imaging (sMRI)**
Individual Project · UGC Theme-Based Research (T45-407/19N) Nov. 2024 - Dec. 2024
Leveraged structural MRI (sMRI) scan data with targeted brain area masks to detect neurocognitive disorders and declines
- **Talktive: a Conversational Agent Using Backchannels to Engage Elderly in Speaking**
Main Contributor · UGC Theme-Based Research (T45-407/19N) Nov. 2020 - Aug. 2021
Designed and implemented a spoken conversational agent with backchanneling to enhance elderly engagement in human-computer interaction; Conducted user studies to evaluate system efficacy
- **Cross-Devices Speaker Recognition**
Main contributor · Tsinghua University & Huawei Noah's Ark Lab Dec. 2019 - Sep. 2020
Investigated solutions to the statistical mismatch problem in cross-device speaker recognition
- **Chinese Text-to-Speech System**
Individual final-year project · Jilin University Jan. 2019 - Apr. 2020
Developed a Chinese text-to-speech PC software with a front-end GUI and back-end text-to-speech model
- **Optical Fiber Methane Gas Sensor**
Student Leader · National Key R&D Program (No. 2016YFC0303902) May. 2017 - Apr. 2018
Developed a gas sensor based on tunable diode laser absorption spectroscopy, and investigated state-of-the-art signal processing algorithms for optical sensors
- **High-speed Time-interleaved ADCs based on FPGA**
Individual summer project · Jilin University Jul. 2017 - Aug. 2018
Proposed a distributed read/write algorithm for ADCs on FPGA chips

AWARDS AND HONORS

- ICML Gold Reviewer 2026
- The Outstanding Student Award of The Chinese University of Hong Kong 2025
- IEEE Ganesh N. Ramaswamy Memorial Student Grant 2025
- ISCA Award for the Best Paper Published in Speech Communication (2019-2023) 2024
- The second place of 2022 ICASSP M2MeT challenge (Track 1) 2022
- Top Cited Article of Speech Communication 2022
- The Chinese University of Hong Kong Postgraduate Studentship 2021
- Jilin University Scholarship 2018
- Jilin University Individual Scholarship 2017

CHALLENGE

- Ranked 2/14 · 2022 ICASSP M2MeT challenge Track 1

TEACHING

- Guest Lecturer · SEEM2460 Introduction to Data Science · The Chinese University of Hong Kong (2026 Spring)
- Teaching assistant · SEEM2460 Introduction to Data Science, The Chinese University of Hong Kong (2025 Spring, 2024 Spring, 2023 Spring, 2022 Spring)
- Teaching assistant · ENGG2780 Statistics for Engineers, The Chinese University of Hong Kong (2023 Spring, 2022 Spring)
- Undergraduate Student Mentor · Department of SEEM, The Chinese University of Hong Kong (2024 Fall)

PATENT

- Method and System for Text-To-Speech Hong Kong Short-Term Patent

PAPERS

Primary Works

1. **Jiawen Kang**, Dongrui Han, Xixin Wu, Helen Meng. "Reading Cognition as Decisions Unfold in Words: A Factorized Inverse Decision Model", *Preprint. arXiv:2608.09222*.
2. **Jiawen Kang**, Kun Li, Xixin Wu, Helen Meng. "Self-Maintained Memory Enhances Faithfulness in RL-based Clinical Agents", *Submitted to NeurIPS 2026*
3. **Jiawen Kang**, Jinchao Li, Mingyu Cui, Junan Li, Xixin Wu, Helen Meng. "TrajLift: Encoding Verbal Memory Dynamics via Heat Diffusion on Semantic Hierarchies", *Submitted to NeurIPS 2026*
4. **Jiawen Kang**, Kun Li, Dongrui Han, Jinchao Li, Junan Li, Lingwei Meng, Xixin Wu, Helen Meng. "Agentic Cognitive Profiling: Realigning Automated Alzheimer's Disease Detection with Clinical Construct Validity", *ICLR 2026 Lifelong Agents Workshop*
5. **Jiawen Kang**, Dongrui Han, Lingwei Meng, Jingyan Zhou, Jinchao Li, Xixin Wu, Helen Meng. "Towards Within-class Variation in Alzheimer's Disease Detection", *INTERSPEECH 2025*
6. **Jiawen Kang**, Lingwei Meng, Mingyu Cui, Yuejiao Wang, Xixin Wu, Xunying Liu, Helen Meng. "Disentangling Speakers in Multi-Talker Speech Recognition with Speaker-Aware CTC", *In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1-5. IEEE, 2025. 🏆 **The recipient of the IEEE 2025 Ganesh N. Ramaswamy Memorial Student Grant**
7. **Jiawen Kang**, Junan Li, Jinchao Li, Xixin Wu, Helen Meng. "Not All Errors Are Equal: Investigation of Speech Recognition Errors in Alzheimer's Disease Detection", *In 2024 IEEE 14th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, pp. 631-635. IEEE, 2024.

8. **Jiawen Kang**, Lingwei Meng, Mingyu Cui, Haohan Guo, Xixin Wu, Xunying Liu, Helen Meng. "Cross-Speaker Encoding Network for Multi-Talker Speech Recognition." *In ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 11986-11990. *IEEE*, 2024.
9. **Jiawen Kang**, Ruiqi Liu, Lantian Li, Yunqi Cai, Dong Wang, Thomas Fang Zheng. "Domain-invariant speaker vector projection by model-agnostic meta-learning." *Proc. Interspeech 2020*, 3825-3829, doi: 10.21437/Interspeech.2020-2562.
10. Lantian Li*, Ruiqi Liu*, **Jiawen Kang***, Yue Fan, Hao Cui, Yunqi Cai, Ravichander Vippperla, Thomas Fang Zheng, Dong Wang. "Cn-celeb: multi-genre speaker recognition." *Speech Communication* 137 (2022): 77-91. (* Equal contribution) 🏆 **ISCA Award for the Best Paper Published in Speech Communication (2019-2023)**
11. Jinchao Li*, Yuejiao Wang*, Junan Li*, **Jiawen Kang***, Bo Zheng, Simon Wong, Manwai Mak, Brian Mak, Xixin Wu, Xunying Liu, Helen Meng. "Detecting Neurocognitive Disorders through Analyses of Topic Evolution and Cross-modal Consistency in Visual-Stimulated Narratives", *IEEE Journal of Selected Topics in Signal Processing* (* Equal contribution)
12. Helen Meng, Brian Mak, Man-Wai Mak, Helene Fung, Xianmin Gong, Timothy Kwok, Xunying Liu, Vincent Mok, Patrick Wong, Jean Woo, Xixin Wu, Ka Ho Wong, Sean Shensheng Xu, Naijun Zheng, Ranzo Huang, **Jiawen Kang ¶**, Xiaoquan Ke, Junan Li, Jinchao Li, Yi Wang. "Integrated and enhanced pipeline system to support spoken language analytics for screening neurocognitive disorders." *Proc. INTERSPEECH 2023*, 1713-1717, doi: 10.21437/Interspeech.2023-2249. (¶ **Student lead; student authors listed alphabetically**)

Technical Report

13. Covo-Audio Team. "Covo-Audio Technical Report", <https://arxiv.org/abs/2602.09823>. (**Core contributor**)

Collaborative Works

14. Lingwei Meng, **Jiawen Kang**, Yuejiao Wang, Zengrui Jin, Xixin Wu, Xunying Liu, Helen Meng. "Empowering whisper as a joint multi-talker and target-talker speech recognition system", *Proc. Interspeech 2024*, 4653-4657, doi: 10.21437/Interspeech.2024-971
15. Mingyu Cui, **Jiawen Kang**, Jiajun Deng, Xi Yin, Yutao Xie, Xie Chen, Xunying Liu. "Towards Effective and Compact Contextual Representation for Conformer Transducer Speech Recognition Systems." *Proc. INTERSPEECH 2023*, 2223-2227, doi: 10.21437/Interspeech.2023-552.
16. Lingwei Meng, **Jiawen Kang**, Mingyu Cui, Haibin Wu, Xixin Wu, Helen Meng. "Unified Modeling of Multi-Talker Overlapped Speech Recognition and Diarization with a Sidecar Separator." *Proc. INTERSPEECH 2023*, 3467-3471, doi: 10.21437/Interspeech.2023-1422.
17. Lingwei Meng, **Jiawen Kang**, Mingyu Cui, Yuejiao Wang, Xixin Wu, Helen Meng. "A Sidecar Separator Can Convert A Single-Talker Speech Recognition System to A Multi-Talker One." *In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1-5. *IEEE*, 2023.
18. Haibin Wu, **Jiawen Kang**, Lingwei Meng, Helen Meng, Hung-yi Lee. "The defender's perspective on automatic speaker verification: An overview." *In IJCAI 2023 DADA Workshop*.

19. Haibin Wu, **Jiawen Kang**, Lingwei Meng, Yang Zhang, Xixin Wu, Zhiyong Wu, Hung-yi Lee, Helen Meng. "Tackling spoofing-aware speaker verification with multi-model fusion." *Proc. The Speaker and Language Recognition Workshop (Odyssey 2022)*, 92-99, doi: 10.21437/Odyssey.2022-13.
20. Zijian Ding, **Jiawen Kang**, Tinky Oi Ting Ho, Ka Ho Wong, Helene H Fung, Helen Meng, Xiaojuan Ma. "TalkTive: a conversational agent using backchannels to engage older adults in neurocognitive disorders screening." *In Proceedings of the 2022 CHI conference on human factors in computing systems*, pp. 1-19. 2022.
21. Yue Fan, **Jiawen Kang**, Lantian Li, KC Li, HL Chen, ST Cheng, PY Zhang, ZY Zhou, YQ Cai, Dong Wang. "Cn-celeb: a challenging chinese speaker recognition dataset." *In ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 7604-7608. IEEE, 2020.
22. Tianyu Zhang, **Jiawen Kang**, Dezhuang Meng, Hongwei Wang, Zhengming Mu, Meng Zhou, Xiaotong Zhang, Chen Chen. "Mathematical Methods and Algorithms for Improving Near-Infrared Tunable Diode-Laser Absorption Spectroscopy." *Sensors* 18, no. 12 (2018): 4295.
23. Lantian Li, Dong Wang, **Jiawen Kang**, Renyu Wang, Jing Wu, Zhendong Gao, Xiao Chen. "A principle solution for enroll-test mismatch in speaker recognition." *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 30 (2022): 443-455.
24. Haohan Guo, Fenglong Xie, **Jiawen Kang**, Yujia Xiao, Xixin Wu, Helen Meng. "QS-TTS: Towards Semi-Supervised Text-to-Speech Synthesis via Vector-Quantized Self-Supervised Speech Representation Learning." *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2024).
25. Lingwei Meng, Shujie Hu, **Jiawen Kang**, Zhaoqing Li, Yuejiao Wang, Wenxuan Wu, Xixin Wu, Xunying Liu, Helen Meng. "Large Language Model Can Transcribe Speech in Multi-Talker Scenarios with Versatile Instructions", *In ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1-5. IEEE, 2025.
26. Dongrui Han, Mingyu Cui, **Jiawen Kang**, Xixin Wu, Xunying Liu, Helen Meng. "Improving Grapheme-to-Phoneme Conversion through In-Context Knowledge Retrieval with Large Language Models", *In 2024 IEEE 14th International Symposium on Chinese Spoken Language Processing (ISCSLP)*, pp. 631-635. IEEE, 2024.
27. Haibin Wu, Lingwei Meng, **Jiawen Kang**, Jinchao Li, Xu Li, Xixin Wu, Hung-yi Lee, Helen Meng. "Spoofing-Aware Speaker Verification by Multi-Level Fusion." *Proc. Interspeech 2022*, 4357-4361, doi: 10.21437/Interspeech.2022-920.
28. Lantian Li, Yang Zhang, **Jiawen Kang**, Thomas Fang Zheng, Dong Wang. "Squeezing value of cross-domain labels: A decoupled scoring approach for speaker verification." *In ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 5829-5833. IEEE, 2021.
29. Jingyan Zhou, Kun Li, Junan Li, **Jiawen Kang**, Minda Hu, Xixin Wu, Helen Meng. "Purple-teaming LLMs with Adversarial Defender Training." *Submitted to Transactions of the Association for Computational Linguistics*.

30. Mingyu Cui, Yifan Yang, Jiajun Deng, **Jiawen Kang**, Shujie Hu, Tianzi Wang, Zhaoqing Li, Shiliang Zhang, Xie Chen, Xunying Liu. "Exploring SSL Discrete Speech Features for Zipformer-based Contextual ASR", *ICASSP 2026*
31. Naijun Zheng, Na Li, Xixin Wu, Lingwei Meng, **Jiawen Kang**, Haibin Wu, Chao Weng, Dan Su, Helen Meng. "The cuhk-tencent speaker diarization system for the icassp 2022 multi-channel multi-party meeting transcription challenge." *In ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 9161-9165. *IEEE*, 2022.
32. Mingyu Cui, Mengzhe Geng, Jiajun Deng, Chengxi Deng, **Jiawen Kang**, Shujie Hu, Guinan Li, Tianzi Wang, Zhaoqing Li, Xie Chen, Xunying Liu. "Exploring Cross-Utterance Speech Contexts for Conformer-Transducer Speech Recognition Systems" *IEEE/ACM Transactions on Audio, Speech, and Language Processing*